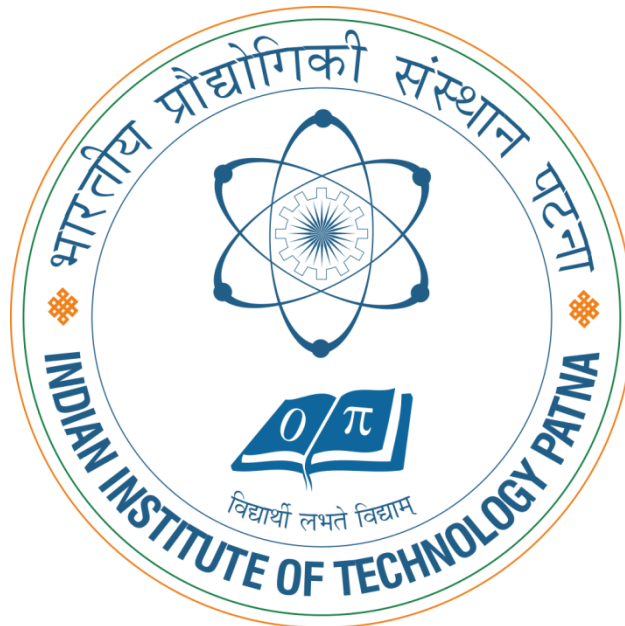


A project report on

Design and optimisation of security gate made for the purpose of
installation at **Reserve Bank of India, Patna**



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Date:-

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Yours faithfully

Souvik Poddar

Kumar Ashish

Mohit Sharma

CHAPTER 1

Security gate-a brief overview

1.1 The need for automated security gates

Security gates are an essential component of industrial and restricted-access facilities. The purpose of an automatic security gate is to ensure that access to specified grounds is limited to authorized personnel. In the majority of applicable settings, industrial-grade gates are placed at the entrance, where access is granted only to authorized vehicles. Depending on the size or range of a given facility, there might also be secondary security gates inside the premises to limit access in certain areas. An example would be any parking garage where all drivers enter in through a main gate, but where employees proceed to a second inner gate to park in reserved spaces that are separate from those allotted to the general public.

Applications:-

- Airports and seaports
- Industrial/commercial factories
- Military bases and border stations
- Water treatment and chemical plants
- Railroad depots, crossings and train yards
- Municipal authorities and transportation terminals

Automatic security gate systems are also used in certain residential settings, including the entrances of exclusive condominium complexes and private neighbourhoods . However, unlike the purpose gates serve at the entrances to industrial facilities — where gates are used to restrict access to ships, planes, weapons, machinery and testing plants — gates in residential settings are used simply to restrict access to residents, family members and guests, while keeping out loiterers, vandals and burglars.

1.2 Components

Gate-

- physical object that is moved to block the gate opening
- made of either ornamental iron or chain-link material

Gate operator

- The gate operator is the machinery that moves the gate in and out of the gate opening
- Gate operators are electrically-powered and may be chain-driven, gear-driven, or hydraulic

1.3 Types of gates

- Slide gate
- Cantilever gate
- Swing gate
- Vertical lift gate
- Vertical pivot lift gate
- Bi-folding gate
- Barrier arm gate

1.3.1 Sliding gate



Fig.1(Sliding gate)

Advantages

1. If space is less or driveway behind the gates is on a steep incline, it is the best possible option
2. Also used for cantilever gating systems

Disadvantages

1. Rollers can be a source of friction
2. There can be problems with the rollers getting blocked by ice or debris

1.3.2 Cantilever gate



Fig 2(Cantilever gate)

The cantilever gate is similar to the slide gate, but does not use rollers that slide along the ground to support it. It is supported by rails running on the inside of the fence structure.

Advantages

1. Since no rollers are used, there is no friction and source of obstruction, so its more reliable.

Disadvantages

1. Since for support the cantilever is made of higher width, it consumes more space and can be pose problems at places where space is limited

1.3.3 Swing gate



Fig 3(Swing gate)

They are hinged on one side and open on the other such that it can swing open and close like a door (fig 3)

Advantages

1. Low cost
2. Ease of installation

Disadvantages

1. People may gate hit or trapped while opening or closing of the gate
2. The mouth of the opening should be sufficient for the vechiles to pass through

1.3.4 Vertical lift gate



Fig 4(Vertical lift gate)

Vertical lift gates move up and down vertically over the gate opening. This type of gate requires that tall vertical support towers be installed on each side of the gate opening (fig 4).

Advantages

1. Ideal when limited space is available next to gate opening

1.3.5 Vertical pivot lift gate



Fig 5(Vertical pivot lift gate)

Vertical pivot lift gates rotate in and out of the gate opening. Vertical pivot lift gates are supported entirely from the gate operator itself and do not require any additional support structures (fig 5).

Advantages

1. Less obtrusive than vertical lift gates due to absence of vertical towers

Disadvantages

1. Requires additional space beside the gate
2. Uses springs as counterweight making them less reliable compared to vertical lift gate

1.3.6 Bi-folding gate



Fig 6(Bi-folding gate)

Bi-folding gates consist of two gate panels that are hinged together. When activated, these gate panels fold back onto themselves to allow access (fig 6)

Advantages

1. When the drive has an incline rising upward into the drive onto which a standard gate would bottom out, the bi-fold design only takes a quarter of the space
2. Requires considerably less power to operate

Disadvantages

1. Additional safety considerations are required because of the presence of many potential entrapment points.

1.3.7 Barrier arm gate



Fig 7(Barrier arm gate)

Barrier arm gates consist of a vertical barrier arm that is rotated in and out of the gate opening. Barrier arm gates are used to control vehicles, not pedestrians (fig 7).

Advantages

1. Barrier arm gates are used primarily to control access in and out of parking facilities, or to control vehicular traffic at manned security entrances

Disadvantages

1. It is very easy for a person to walk beside or climb over or under the gate arm , barrier arm gates provide almost no security

1.3.8 Tiger teeth gate



Fig 8(Tiger teeth gate)

Tiger Tooth Traffic controllers enforce “One Way” traffic flow for parking facilities or other single direction vehicle lanes. The torsion spring actuated Tiger Teeth positively control traffic in one direction while permitting free flow in the other direction. The Tiger Teeth will puncture all pneumatic tires including steel belted radial and large truck tires (fig 8).

CHAPTER 2

Project purpose

The purpose of our project is to design and develop a model of a security gate to be installed at the Reserve Bank of India, Patna and at the same time make a small prototype of the original gate and make it fully automated.

Hardship for installation of the gate is mainly the excessive water accumulation below the gate due to rainfall which occurs due to poor drainage system in Patna. Also the space available for installation of the gate is too less.

The concept behind our project is to develop a sliding gate which as a whole would not be operated in case to be opened. The gate is divided into two parts and each part can be operated at a time for running purpose. The gate is designed such that for small objects only part of the gate will be operated and for larger objects other part of the gate will be operated. In this way we can reduce the overhang that occurs.

CHAPTER 3

Mechanical design

We have used Creo and solidworks for making all the parts of the gate. The assembly has been made in creo . All the dimensions are taken in mm.

3.1 PTC creo



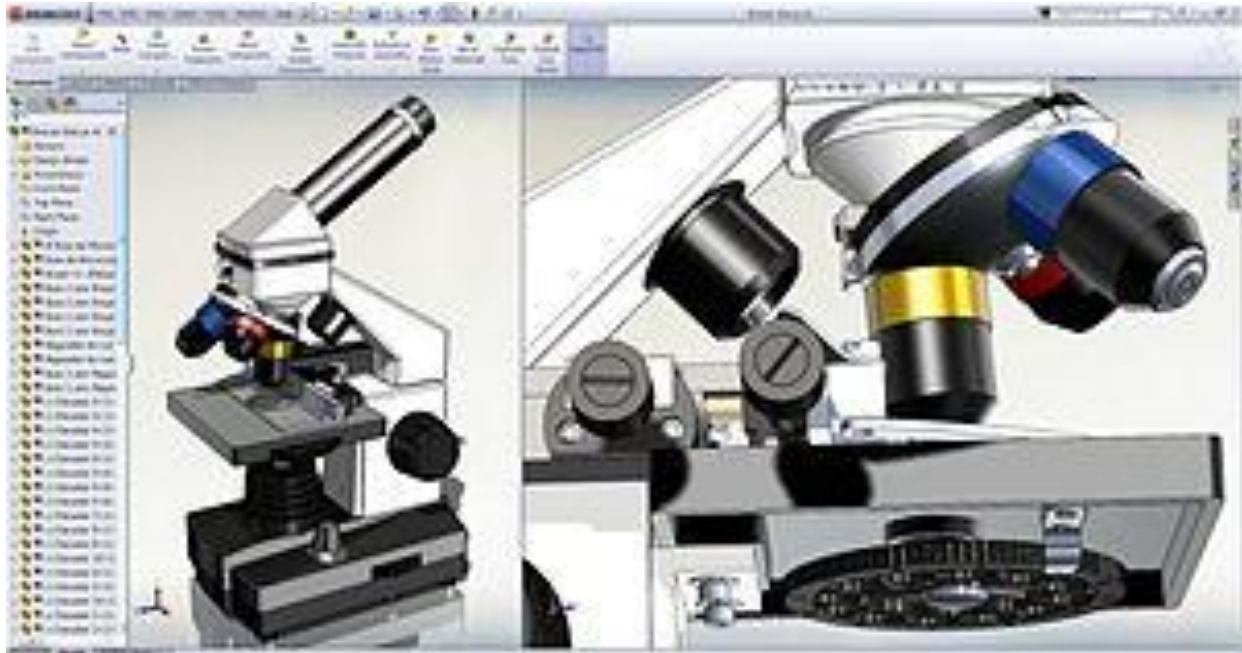
creo®

Creo is a family or suite of design software supporting product design for discrete manufacturers and is developed by PTC. The suite consists of apps, each delivering a distinct set of capabilities for a user role within product development.

Creo Elements/Pro and **Creo Parametric** compete directly with CATIA, Siemens NX/ Solidedge , and Solidworks . The Creo suite of apps replace and supersede PTC's products formerly known as Pro/ENGINEER, CoCreate, and ProductView. Creo has many different software package solutions and features. Creo Illustrate is a good example.

Creo is part of a broader product development system developed by PTC. It connects to PTC's other solutions that aid product development, including Windchill for Product Lifecycle Management (PLM), Mathcad for engineering calculations and Arbortext for enterprise publishing software.

3.2 Solidworks



SolidWorks (stylized as **SOLIDWORKS**) is a solid modeling computer-aided design (CAD) and computer-aided engineering (CAE) computer program that runs on Microsoft Windows. SolidWorks is published by Dassault Systèmes.

SolidWorks Corporation was founded in December 1993 by Massachusetts Institute of Technology graduate Jon Hirschtick. Hirschtick used \$1 million he had made while a member of the MIT Blackjack Team to set up the company. Initially based in Waltham, Massachusetts, United States, Hirschtick recruited a team of engineers with the goal of building 3D CAD software that was easy-to-use, affordable, and available on the Windows desktop. Operating later from Concord, Massachusetts, SolidWorks released its first product *SolidWorks 95*, in November 1995. In 1997 Dassault, best known for its CATIA CAD software, acquired SolidWorks for \$310 million in stock. Jon Hirschtick stayed on board for the next 14 years in various roles. Under his leadership, SolidWorks grew to a \$100 million revenue company.

3.3 CAD Model of different parts

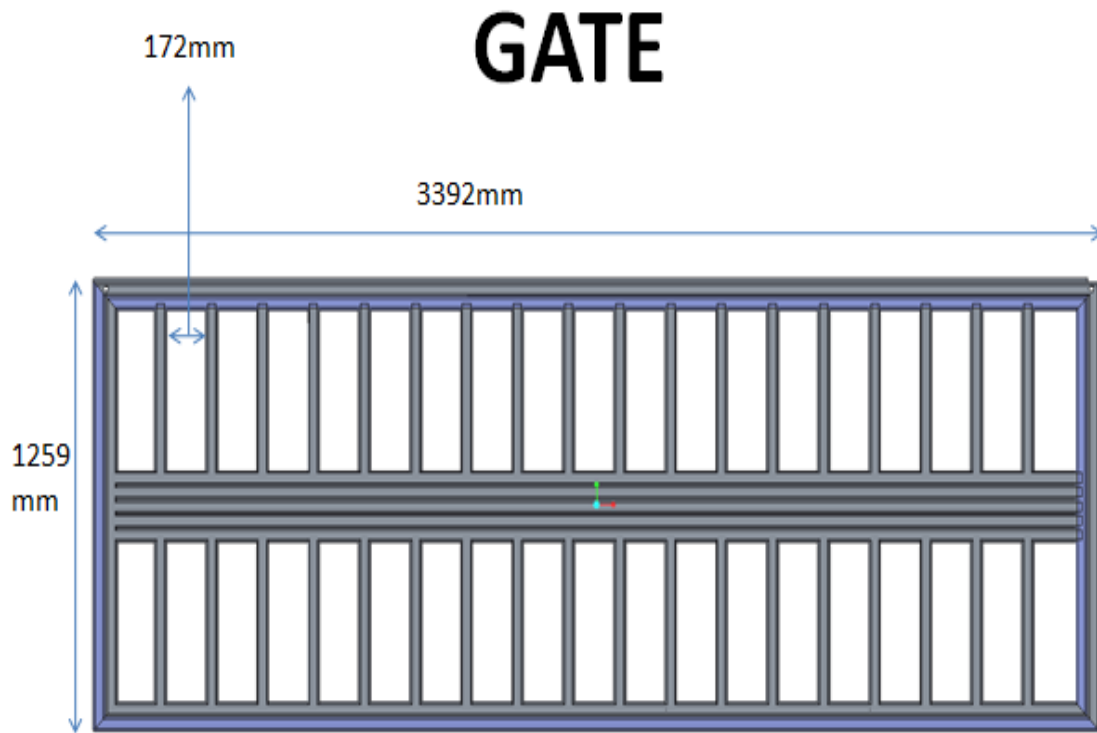


Fig 9(Gate CAD model)

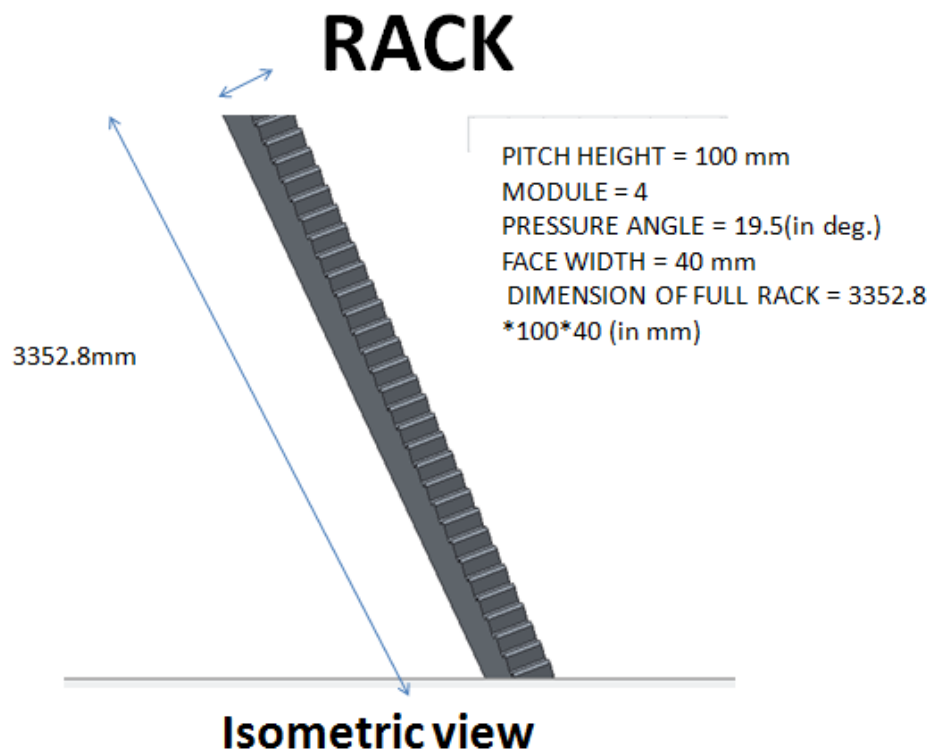


Fig 9(Rack CAD model)

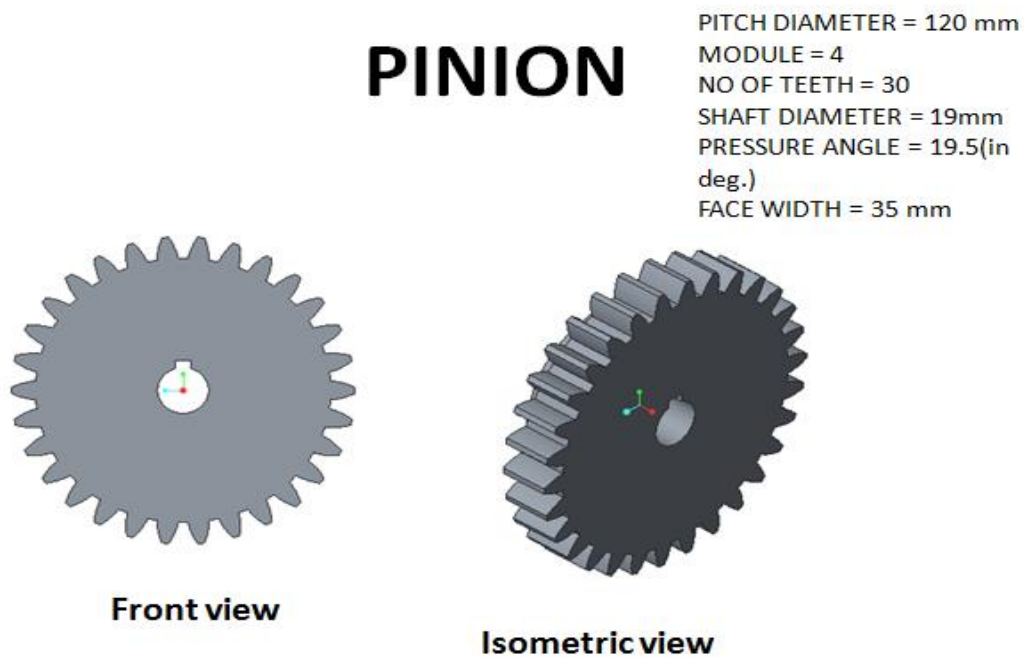


Fig 10(Pinion CAD model)

3.4 Whole assembly of the gate

Closed gate different views

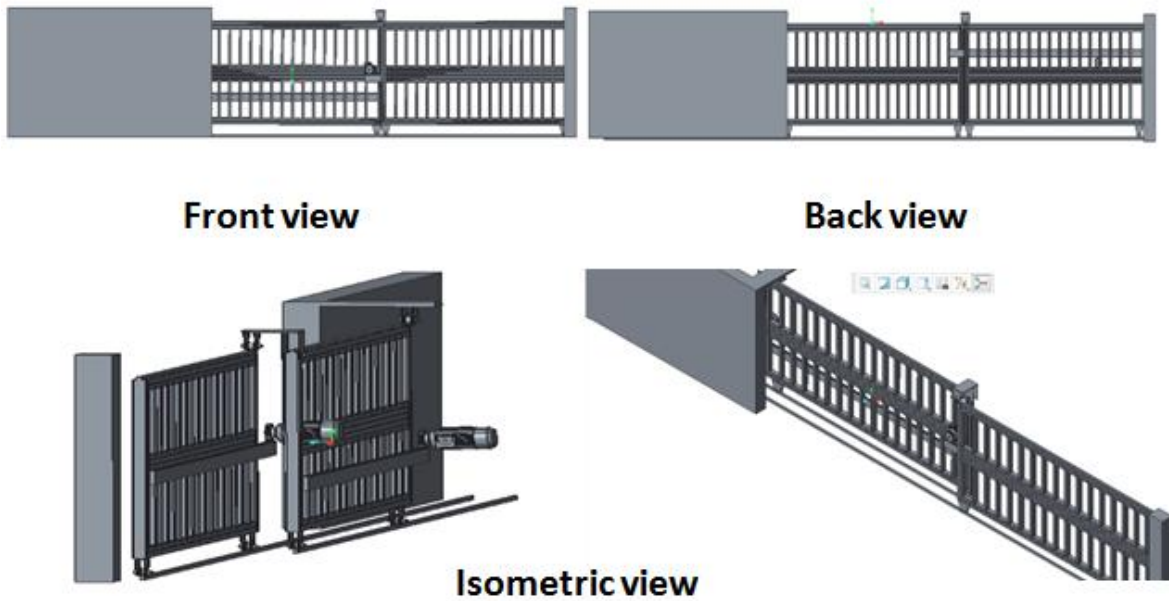


Fig 11(Gate when closed)

Half open gate different views

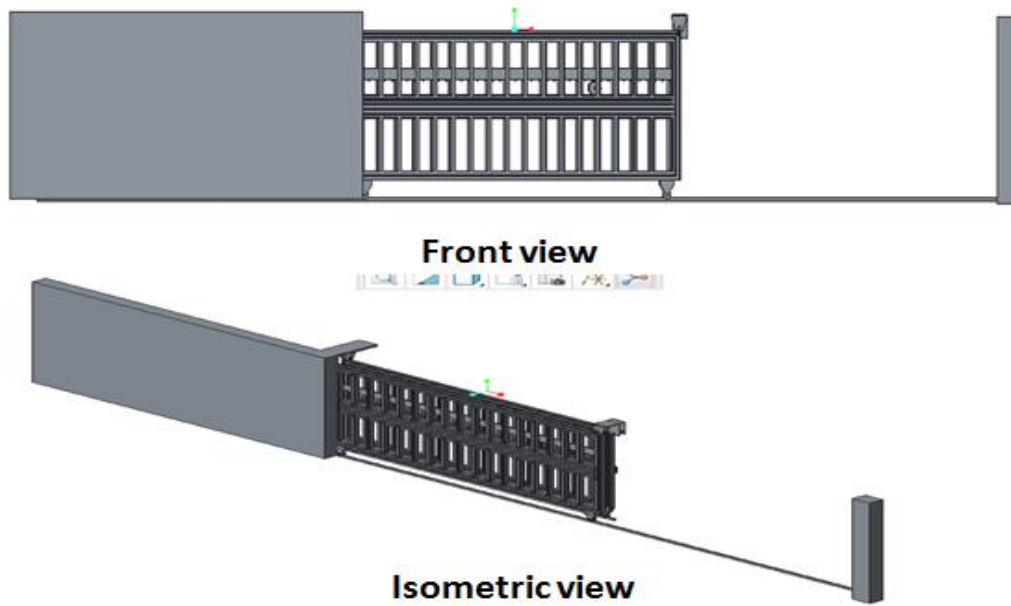


Fig 12(Gate when half opened)

Open gate different views

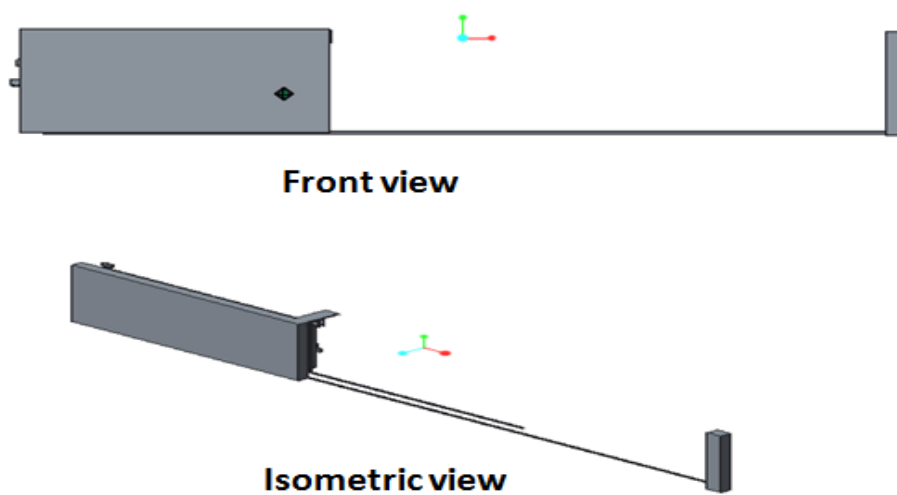
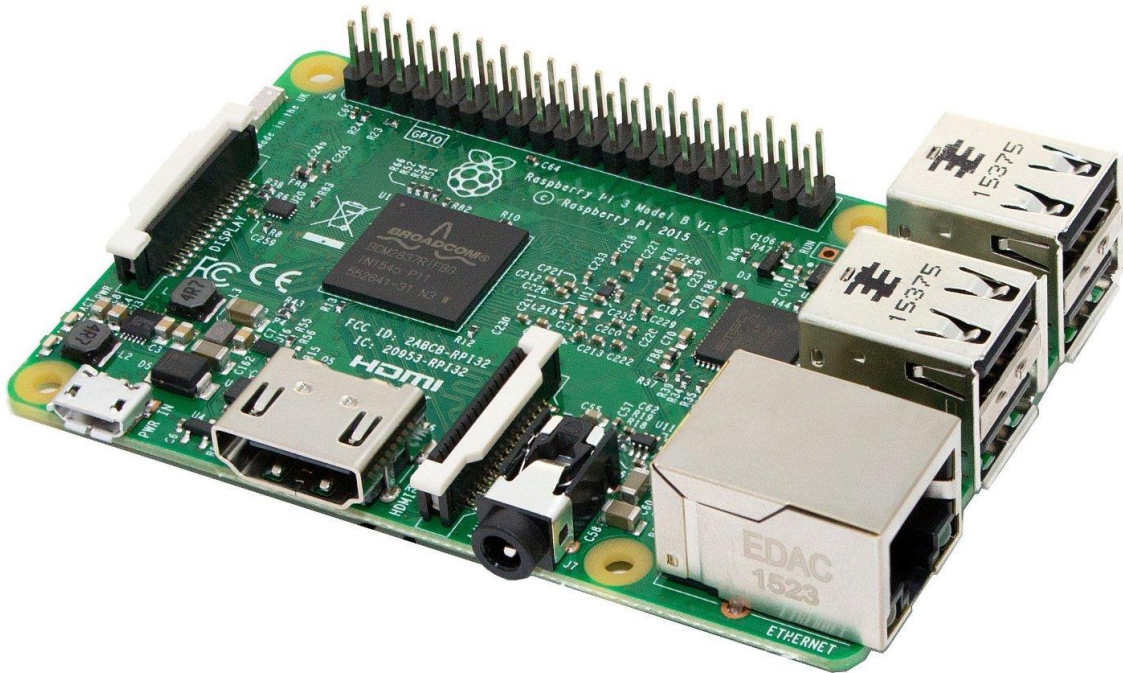


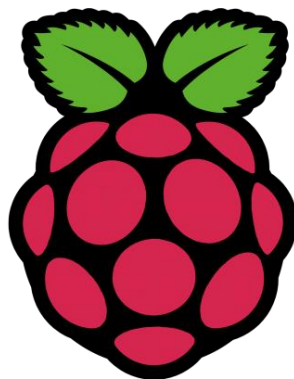
Fig 13(Gate when fully opened)

CHAPTER 4

Controller design



4.1 Raspberry pi 3



In this project we have used raspberry pi as our microcontroller and we have interfaced it with a android phone where we have made buttons and by the click of buttons the gate will be operated.

The Raspberry Pi is a small, barebones computer developed by The Raspberry Pi Foundation, a UK charity, with the intention of providing low-cost computers and free software to students. Their ultimate goal is to foster computer science education and they hope that this small, affordable computer will be a tool that enables that.

The printed circuit board (PCB) houses the input and output connectors as well as the computer hardware itself. Currently, the Foundation is selling a naked PCB — meaning there is no included Raspberry Pi case — and will release a cheaper version, with fewer connectivity options, soon. These two versions without cases are essentially the beta run and buildup to the release of the final product. The final version will be an educational edition with case, documentation, and pre-loaded educational software. On the software side of things, there are currently three Linux-based operating systems supported by the Raspberry Pi.

There were 10,000 Model B versions produced in the first production batch, and they all sold out within hours of going on sale. The Model B is the \$35 version with the most connectivity options, and is the model that most enthusiasts are interested in. A Model A without ethernet and with a single USB port is on the way and will be sold for \$25. Furthermore, the final educational edition Raspberry Pi computers (with case and extras) are slated for a summer 2012 release, though the price is unknown.

4.1.1 Specifications and performance

As for the specifications, the Raspberry Pi is a credit card-sized computer powered by the Broadcom BCM2835 system-on-a-chip (SoC). This SoC includes a 32-bit ARM1176JZFS processor, clocked at 700MHz, and a Videocore IV GPU. It also has 256MB of RAM in a POP package above the SoC. The Raspberry Pi is powered by a 5V micro USB AC charger or at least 4 AA batteries (with a bit of hacking).

While the ARM CPU delivers real-world performance similar to that of a 300MHz Pentium 2, the Broadcom GPU is a very capable graphics core capable of hardware decoding several high definition video formats. However, in order to keep costs of the Raspberry Pi low, the UK charity has only licensed the H.264 codec for hardware decoding (and it is unclear if users will be able to purchase/activate additional codecs). In that regard, the Videocore IV GPU is rather potent as it is capable of hardware decoding 1080p30 H.264 with bit-rates up to 40Mb/s.

4.1.2 What does the Raspberry Pi do?

The allure of the Raspberry Pi comes from a combination of the computer's small size and affordable price. Enthusiasts envision using the small form-factor PC as a cheap home theater PC (HTPC), or secondary low-power desktop. Institutions, like schools and businesses, could benefit from deploying a fleet of computers for a fraction of the cost of traditional desktop towers. The small size makes for an easy-to-hide computer that sips power and can be mounted behind the display with an appropriate case. It could also be used in niche applications, like digital signage.

4.1.3 Raspberry pi 3 gpio pins layout

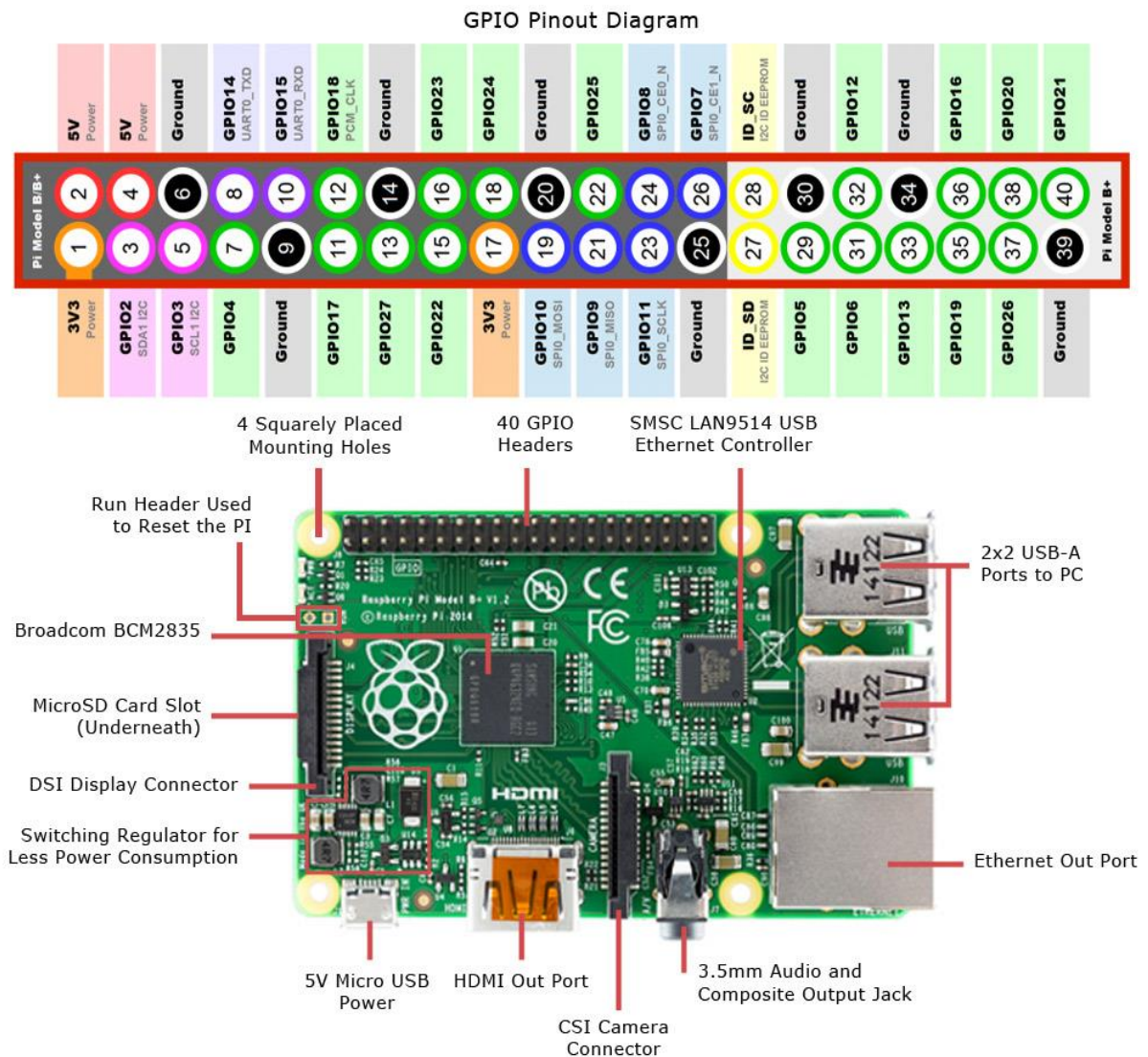


Fig 14(GPIO pins layout)

4.1.4 Operating system

Raspbian

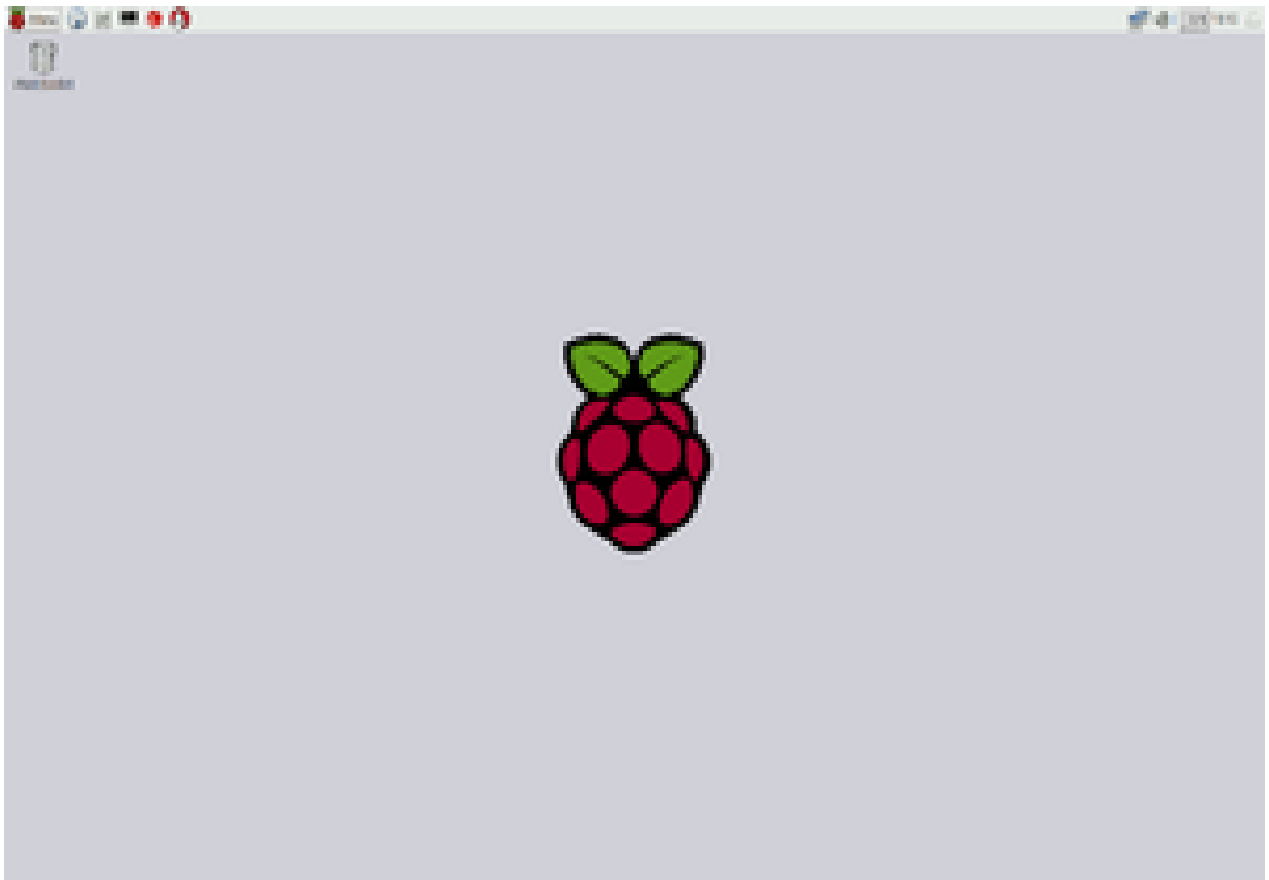


Fig 15(Raspbian Desktop)

Raspbian is a Debian-based computer operating system for Raspberry Pi. Since 2015 till now it is officially provided by the Raspberry Pi Foundation as the primary operating system for the family of Raspberry Pi single-board computers. Raspbian was created by Mike Thompson and Peter Green as an independent project. The initial build was completed in June 2012. The operating system is still under active development. Raspbian is highly optimized for the Raspberry Pi line's low-performance ARM CPUs.

Raspbian uses PIXEL, **Pi Improved Xwindows Environment, Lightweight** as its main desktop environment as of the latest update (fig 15). It is composed of a modified LXDE desktop environment and the Openbox stacking window manager with a new theme and few other changes. The distribution is shipped with a copy of computer algebra program Mathematica and a version of Minecraft called Minecraft Pi as well as a lightweight version of Chromium as of the latest version.

4.2 Coding ground-Python

Python is a wonderful and powerful programming language that's easy to use (easy to read and write) and with Raspberry pi lets you connect your project to the real world.



Python syntax is very clean, with an emphasis on readability and uses standard English keywords .We start by opening IDLE in desktop.

4.2.1 IDLE

The easiest introduction to python is through IDLE, a python development environment. Open IDLE from the Desktop or applications menu(fig 16)

IDLE gives you a REPL (Read-Evaluate-Print-Loop) which is a prompt you can enter python commands in to.

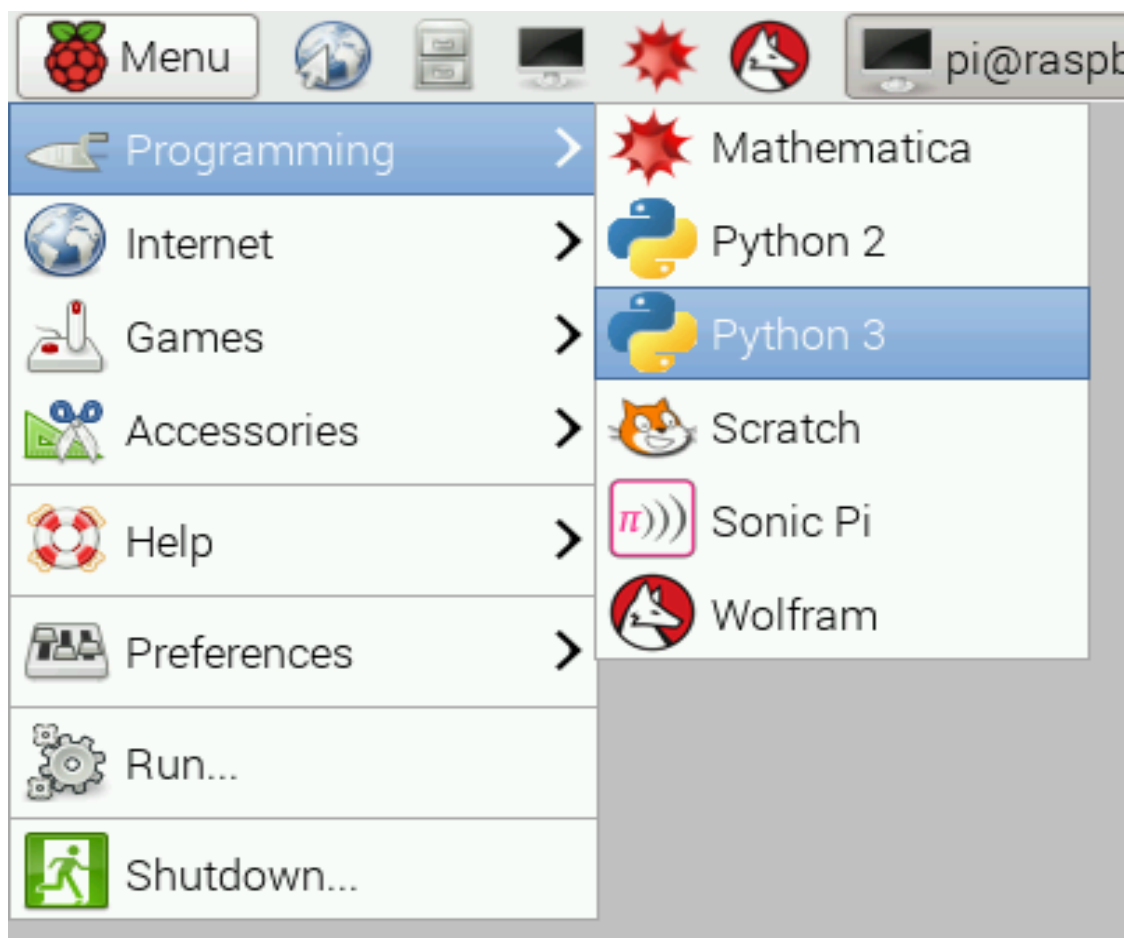


Fig 16

Python has two versions: python 2 and python 3. Python 3 is the newest version and recommended, however python 2 is available for legacy applications which do not support python 3 yet.

4.3 How to make the connections wireless

Follow the sequence:-

Installing and configuring VNC server

```
sudo apt-get install tightvncserver
sudo apt-get install pythontk
sudo vnc server:1
< password>
< verify password>
sudo apt-get realvnc-vnc-server realvnc-vnc-viewer
```

4.4 Giving static IP address

```
sudo nano/etc/dhcpd.conf
```

(In editor panel type the following commands)

```
Interface wlan0
static IP_address=192.168.0.135/24
static routers=192.168.0.1
static domain_name-servers=192.168.0.1
```

4.5 Programming in python

The programming is made such that we can control the motors by clicking of buttons. The details of the programming is attached as a file below. We have made four buttons for this purpose.

1. MOVE GATE 1 FORWARD
2. MOVE GATE 2 BACKWARD
3. MOVE GATE 2 FORWARD
4. MOVE GATE 2 BACKWARD

4.5.1 CODE

```
import tkinter as tk
import tkinter.font
from gpiozero import LED
import time

win=tk.Tk()
win.title("Using tkinter")
myFont=tkinter.font.Font(family = 'Helvetica' ,size = 12,weight = "bold")
led1= LED(21)
led2= LED(20)
led3= LED(26)
led4= LED(16)
def led1Toggle():
    if led1.is_lit:
        led1.off()
        ledButton["text"]="move gate1 forward"
    else:
```

```

        led1.on()
        ledButton["text"]="stop gate1"

def led2Toggle():
    if led2.is_lit:
        led2.off()
        ledButton["text"]="move gate1 forward"
    else:
        led2.on()
        ledButton["text"]="stop gate1"

def led3Toggle():
    if led3.is_lit:
        led3.off()
        ledButton["text"]="move gate1 forward"
    else:
        led3.on()
        ledButton["text"]="stop gate1"

def led4Toggle():
    if led4.is_lit:
        led4.off()
        ledButton["text"]="move gate1 forward"
    else:
        led4.on()
        ledButton["text"]="stop gate1"

def exitProgram():
    win.quit()

ledButton1=tk.Button(win, text='gate1 forward activate' , font=myFont, command=led1Toggle,
bg='bisque', height=1, width=24)
ledButton1.grid(row=0, sticky=tk.NSEW)
ledButton2=tk.Button(win, text='gate1 forward activate' , font=myFont, command=led2Toggle,
bg='bisque', height=1, width=24)
ledButton2.grid(row=2, sticky=tk.NSEW)
ledButton3=tk.Button(win, text='gate2 forward activate' , font=myFont, command=led3Toggle,
bg='bisque', height=1, width=24)
ledButton3.grid(row=4, sticky=tk.NSEW)
ledButton4=tk.Button(win, text='gate2 forward activate' , font=myFont, command=led4Toggle,
bg='bisque', height=1, width=24)
ledButton4.grid(row=6, sticky=tk.NSEW)

tk.mainloop()

```


4.6 For controlling raspberry pi wirelessly through laptop

- Install VNC viewer in laptop
- Install PUTTY
- Connect the raspberry pi and laptop to the same router
- Open vnc server in pi (sudo vnc server:1)
- Give the IP address of raspberry pi in putty and click connect
- Open VNC viewer and give username and password of the pi
- Control the raspberry pi wirelessly through laptop

4.7 For controlling raspberry pi wirelessly through android phone

- Switch on vnc server in raspberry pi
- Download VNC connect in android from playstore
- Give IP address of the raspberry pi in VNC connect and click connect
- The screen of raspberry pi opens in android and we can control it through our phone

4.8 How to find IP address of raspberry pi

- Give power to raspberry pi
- Download advanced IP scanner in laptop
- Search all available devices with their IP address
- You will find a device of raspberrypi foundation. That one is your pi IP address

4.9 Use of relays

Since the output of the GPIO pins are limited 3.3V we need to use relay to drive powerful motors. We have used relay made especially for Raspberry pi. With the use of relays large powerful motors can be driven without any damage to the raspberry pi (fig 17).



Readymade single channel relays

Circuit for driving motor in opposite directions

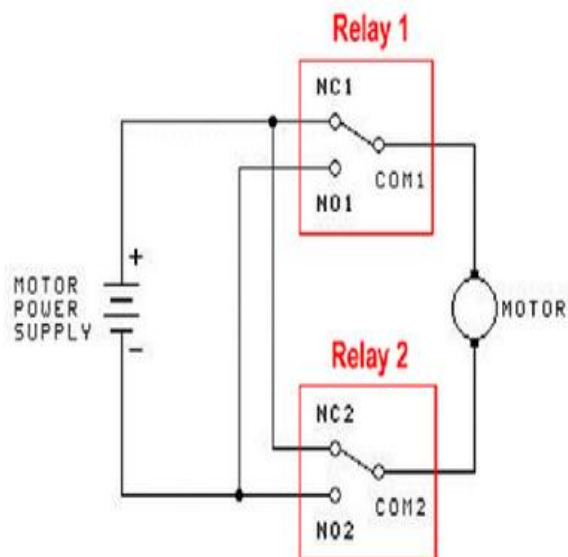


Fig 17(Relay circuit)

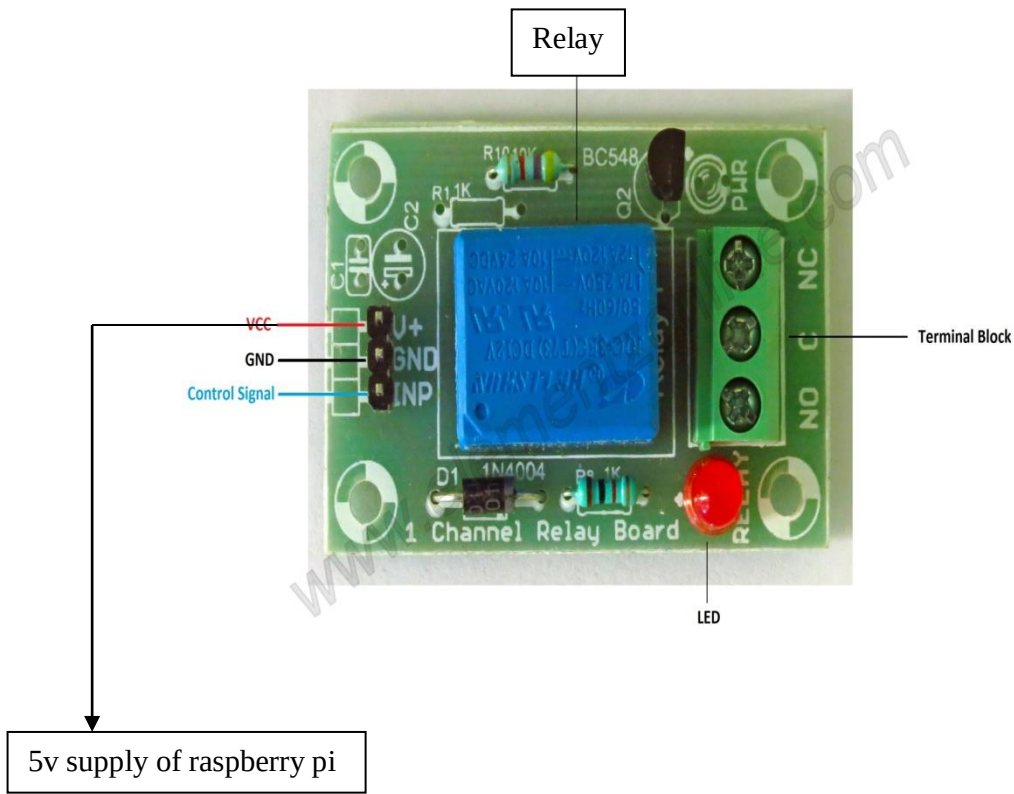


Fig 18(Relay pinout)

4.9.1 Specifications

1. Equipped with high-current relay 7A@250VAC / 10A@24VDC
2. Power source: DC
3. It can control both AC and DC appliances such as Solenoids, Motors, lights, fans, etc.
4. Voltage: 5V
5. Can be controlled directly by a wide range of microcontrollers such as Arduino , AVR, PIC, ARM, 8051 and so on.

CHAPTER 5

Working model

5.1 Mechanical part

In our working model we have acrylic as the material for making all the parts of the gate. We have designed the parts in Corel draw and have cut the materials using laser cutting. The material of the gate body has been made of plywood.

Acrylic and wood has been chosen for giving the model as much light weight as possible.

Density of acrylic=1.18 g/cm³

5.2 Laser cutting

Laser cutting is a technology that uses a laser to cut materials, and is typically used for industrial manufacturing applications, but is also starting to be used by schools, small businesses, and hobbyists. Laser cutting works by directing the output of a high-power laser most commonly through optics. The laser optics and CNC (computer numerical control) are used to direct the material or the laser beam generated. A typical commercial laser for cutting materials would involve a motion control system to follow a CNC or G-code of the pattern to be cut onto the material. The focused laser beam is directed at the material, which then either melts, burns, vaporizes away, or is blown away by a jet of gas, leaving an edge with a high-quality surface finish(fig 19 and fig 20). Industrial laser cutters are used to cut flat-sheet material as well as structural and piping materials.

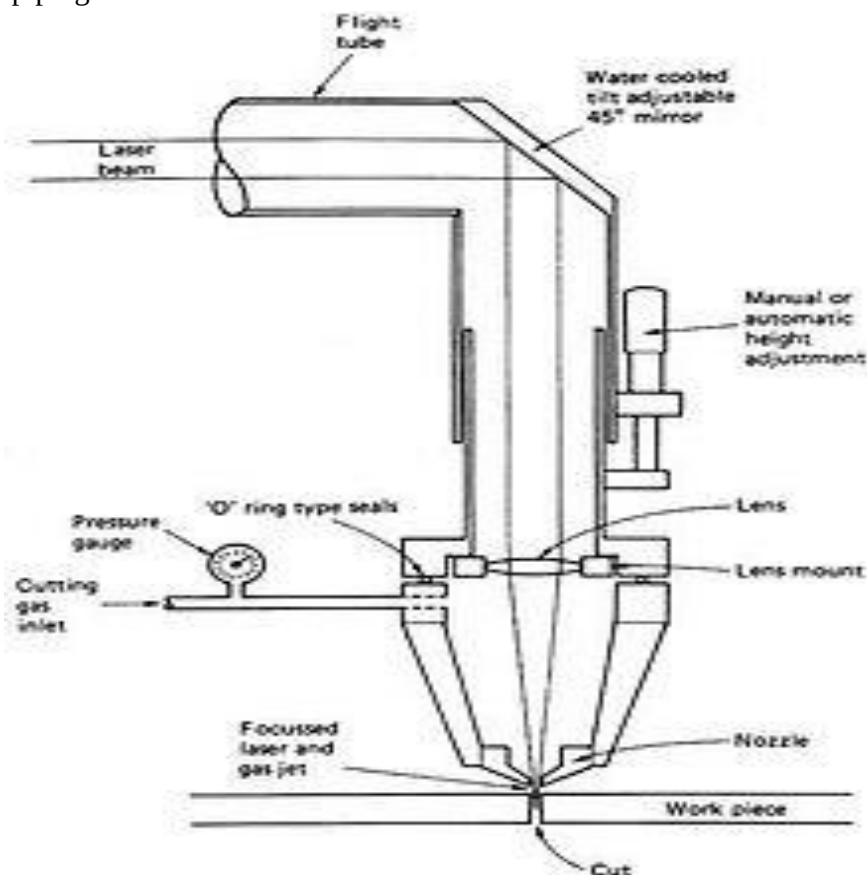


Fig 19(Laser cutting)



Fig 20(Laser cutting operation)

5.3 Mechanical components

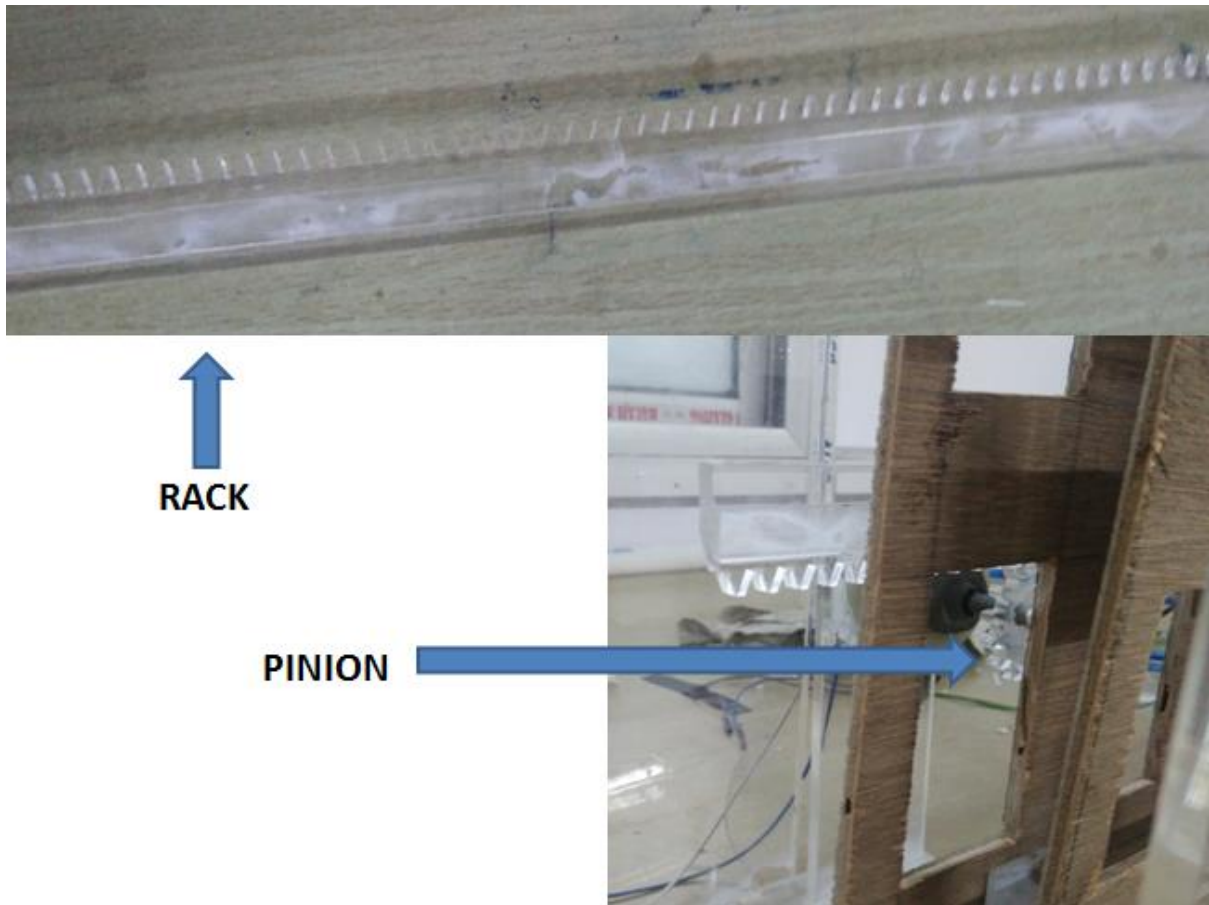


Fig 21(Racks and pinions)

5.3.1 Specifications

Gate

Length =40cm

Breadth=0.5cm

Height= 23.4

Material used = ply wood

Rack

Length =42.5cm

Breadth= 0.5cm

Height= 2.5cm

Tooth thickness=0.15cm

Pinion

Pitch radius = 1.6cm

No of teeth= 15

Outer diameter= 3.65cm

Addendum = 2.25cm

Guide

Length = 120cm

Breadth = 0.3cm

Height = 1cm

Roller

Radius of bigger cylinder= 1.250 cm

Extrude of bigger cylinder= 0.3cm

Radius of smaller cylinder= 0.9cm

Extrude of smaller cylinder= 0.5cm

5.4 Final working model



Fig 22(working model final)

5.5 Controller

We have used raspberry pi as mentioned above as our controller. In case operated manually we can use DPDT switches (fig 23) for our purpose.

5.5.1 DPDT switches



Fig23(DPDTswitches)



Fig24(Control panel)

5.5.2 DPDT connections

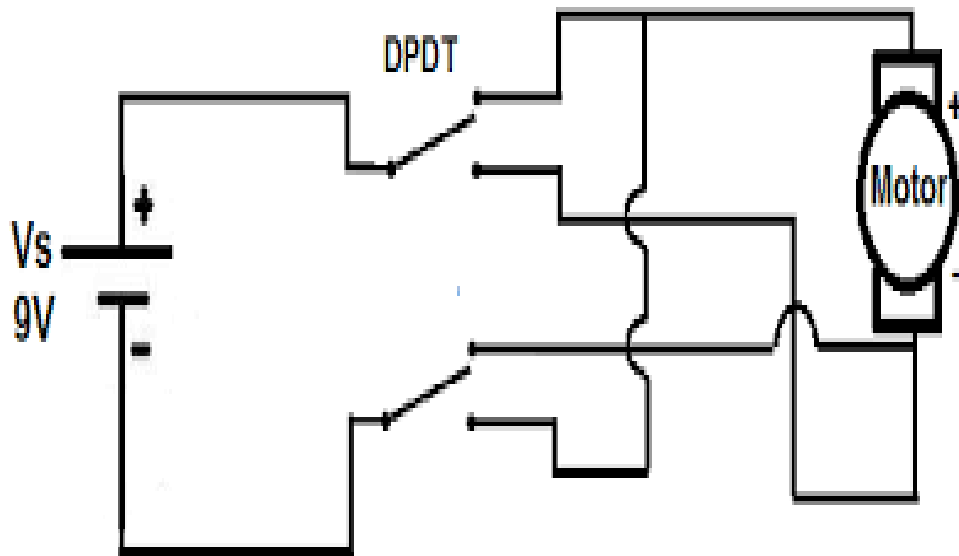


Fig 25(DPDT connection)

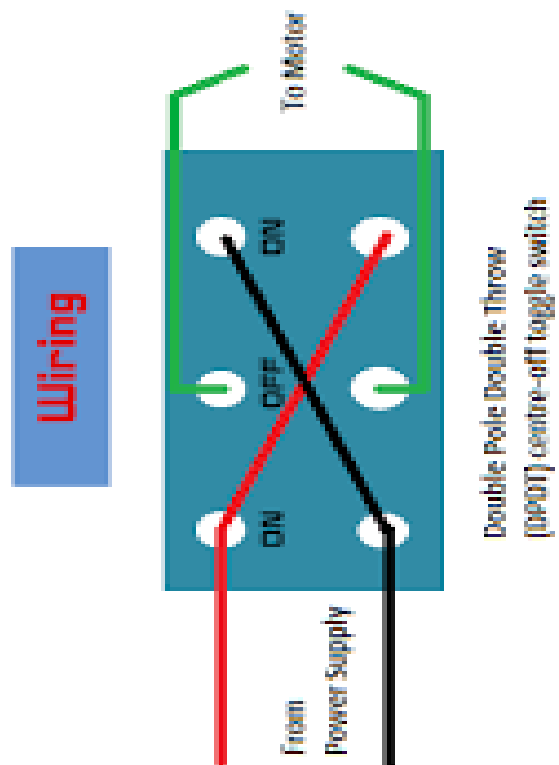


Fig 26(DPDT circuit)

CHAPTER 6

SPECIFICATIONS AND ESTIMATED COST

DIFFERENT COMPONENTS OF SLIDING GATE

6.1 GATE

- 1.1 DIMENSION = 3392*1259*60
- 1.2 MATERIAL USED = AL6061 Rectangular hollow tube
 - 1.2.1 Dimension of tube = 60*40(3.2 thickness) (in mm)
- 1.3 Weight required = 85kg
- 1.4 Cost = INR320/kg (approx.)

Total cost=INR54400

6.2 MOTOR

- 2.1 TYPE = INLINE HELICAL GEARED MOTOR
- 2.2 CAPACITY = 400 kg
- 2.3 TORQUE(max.) = 150 N-m
- 2.4 POWER = 1.5h.p (3 Phase)
- 2.5 OUTPUT R.P.M (max.) = 250
- 2.6 FOOT MOUNTED GEARBOX AND FLANGED MOUNTED MOTOR
- 2.7 SHAFT DIAMETER = 19mm
- 2.8 PRICE = INR19,850/piece (approx.)

Total cost=INR 39700

6.3 PINION

- 3.1 MATERIAL = NYLON 6
- 3.2 PITCH DIAMETER = 120 mm
- 3.3 MODULE = 4
- 3.4 NO OF TEETH = 30
- 3.5 SHAFT DIAMETER = 19mm (with a key)
- 3.6 PRESSURE ANGLE = 19.5(in deg.)
- 3.7 FACE WIDTH = 35 mm
- 3.8 PRICE = INR 2600/ piece (approx.)

Total cost=INR 5200

6.4 RACK

- 4.1 MATERIAL = NYLON 6
- 4.2 PITCH HEIGHT = 100 mm
- 4.3 MODULE = 4
- 4.4 PRESSURE ANGLE = 19.5(in deg.)
- 4.5 FACE WIDTH = 40 mm
- 4.6 DIMENSION OF FULL RACK = 3352.8 *100*40 (in mm)
- 4.7 NO OF RACK REQUIRED = 2
- 4.8 PRICE = INR8900/40MM X 40MM X 1220MM

Total cost=INR 20000

6.5 ROLLER AND GUIDE

- 5.1 DIMENSION = 9144*20*10 & 6096*20*10 (in mm)

6.6 NUTS AND BOLTS

6.1 STANDARD OF BOLT = ISO 14579-8.8(M16)

6.2 DIMENSION OF BOLT = 16mm (diameter) & 50 mm (length)

6.3 DIMENSION OF NUT = as per the dimension of bolt .

6.4NO OF PIECE = 20

6.5 PRICE = INR 10/piece

Total cost=INR 2

00

6.7 ALUMINIUM BAR

7.1 DIMENSION= 6.5*1250*2500 (in mm)

7.2 COST = INR 300/kg (approx.)

TOTAL ESTIMATE EXPENSE FOR THIS PROJECT :-INR1.3lakh

CHAPTER 7

CALCULATIONS

7.1 How to get the specification of parts for the project?

7.1.1 For finding out the torque that is applied on the gate for its motion(fig 27):

- 1) We have to get the weight of the gate for finding out the forces acting on the gate.
- 2) So for that we draw the model of the gate in CAD software by applying the material i.e. aluminium 6061.
- 3) As for why we choose “Aluminium 6061” because its light weight, resistive to rusting, have sufficient material strength and available in India at moderate rate.
- 4) After finding out the weight from the cad model which is around 85kg and further considering the weight of rack, motor, supportive platform we get weight around 110 kg.
- 5) So now we have to find the reaction force which will help us in finding the friction force.
- 6) Moreover the coefficient of friction between steel guideway and steel roller is 0.75.

- 7) So the reaction force comes out

$$\Rightarrow \text{friction coeff.}(u)=0.75$$

$$\Rightarrow \text{Weight}(W)=110*10N$$

$$\Rightarrow W=R1+R2=2*R=1100N$$

$$\Rightarrow \text{Reaction Force}(R)=550N$$

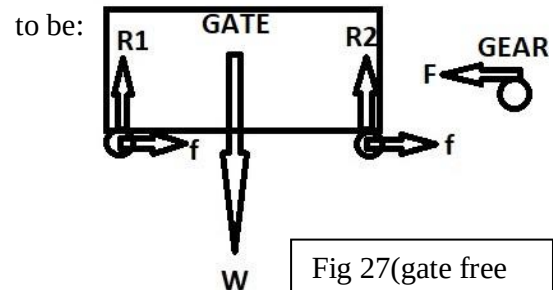


Fig 27(gate free body diagram)

- 8) Therefore the friction force comes out to be:

$$\Rightarrow \text{friction force}(f)=u*R$$

$$\Rightarrow f=0.75*550=412.5N$$

- 9) by using the above eqn we will find out overall force which will help us in finding out the overall torque:

$$\Rightarrow \text{Overall force}(F)=4*f=1650N$$

$$\Rightarrow \text{Torque}(T)=F*r \quad [\text{radius of pinion}(r)=60\text{mm}]$$

$$\Rightarrow T=1650*0.06=99\text{Nm} \approx 100\text{Nm}$$

$$\Rightarrow \text{Safety Factor}(N_{sf})=1.5$$

$$\Rightarrow \text{Torque Required}(T')=T*N_{sf}=100*1.5=150\text{Nm}$$

- 10) So the overall torque required comes out to be 150Nm.

7.1.2 For Finding the suitable motor required to drive the gate:

- 11) As we get the value of torque, so now we can find out the rpm of the motor.
- 12) As most are motors fitted in gate are around 1-3 hp, therefore we selected 1.5 hp motor and thus we do the calculation to verify that the certain motor can satisfy our requirement i.e. gate should be closed in 10-15 sec.
 - $\Rightarrow \text{Power}(P) = 1.5\text{hp} \approx 1000W$
 - $\Rightarrow \text{As Power } (P) = T' * \omega$
 - $\Rightarrow \omega = \frac{\text{Power}(P)}{\text{Torque}(T')} = 1000/150 = 6.67 \text{ rad/s}$
 - $\Rightarrow N = (60*\omega)/2\pi = 63.69 \text{ rpm.}$
 - $\Rightarrow v = r\omega = 0.06*6.67 = 0.4\text{m/s.}$

⇒ So, Time required to fully close the gates = 15 sec.

13) If everything works ideally then safety factor can be reduce to 1 rather than 1.5 , therefore giving us the torque of the gate 100Nm

⇒ $\omega = 1000/100=10\text{rad/s}$

⇒ Then, $v = 0.06*10=0.6\text{m/s}$

⇒ So, Now Time required to fully close the gates = 10 sec.

[NOTE :

We have chosen 1.5 hp motor because if we chose higher power motor then rpm will increase & closing time of gate decreases but due to this, the dimension of rack and gear will change as dynamic force will increase which will increase the cost and weight of overall Gate.]

Therefore,

1.5hp, 150Nm torque and minimum of 70 rpm motor can be used for our project.

7.1.3 For finding the suitable gear for motor 1:

14) So to the find the required gear we select a standard gear which is generally used in sliding gate then we use different strength eqn to verify that the gear we selected is fulfilling our requirements or not.

We take a standard gear having No. of teeth=30 ; Pitch Diameter(D)=120mm=4.72in;Module=4 ; for finding out its face width and verifying that the gear is valid(fig 28).

15) As we know that the overall force is the transmitted force on the motor so we calculate surface speed and dynamic force acting on the gear

⇒ Transmitted force(F_t)=overall force(F)

⇒ $F_t=1650\text{N}=370.9\text{lb}$

[NOTE:

Here we are taking transmitted force=1650N i.e. force equivalent of both the gate combined because we are taking consideration if the motor of gate 2 failed then can the gear on the motor 1 open the gate without breaking]

⇒ Diametral Pitch(P_d)=No. of teeth/Pitch diameter

⇒ $P_d=30/4.72=6.35\text{in}$

⇒ Surface Speed (V_m)= $(\pi * D * N)/12=(3.14*4.72*63.69)/12$

⇒ $V_m=78.7\text{ft/min}$

⇒ Dynamic Force(F_d)= $[(600+V_m)*F_t]/600=(678.7*370.9)/600$

⇒ $F_d=419.55\text{lb}$

16) Now for finding out the width of the gear we use **Lewis equation**i.e.

⇒ $F_s=(S_{al}*b*Y)/P_d$

- F_s =Allowable force
- S_{al} =allowable tensile stress
- b =face width
- Y =Lewis Factor
- P_d =Diametral pitch

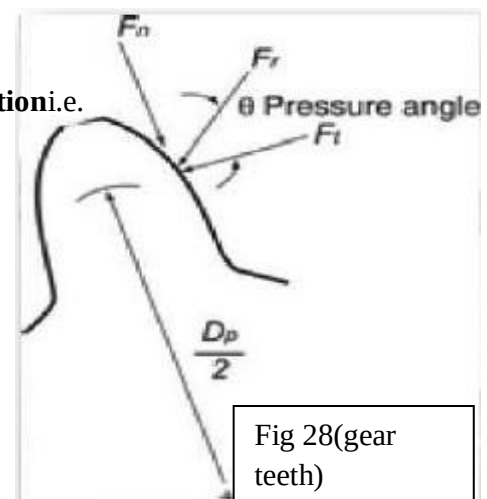


Fig 28(gear teeth)

17) And the relation between F_s and F_d are:

$$\Rightarrow \frac{\text{Allowable force}(F_s)}{\text{Safety Factor}(N_{sf})} \geq \text{Dynamic force}(F_d)$$

18) So the new lewis equation comes out to be

$$\Rightarrow F_d = (S_{al} * b * Y) / (N_{sf} * P_d)$$

19) Therefore the value of face width (b) =

$$\Rightarrow b = \frac{F_d * N_{sf} * P_d}{S_{al} * Y}$$

20) Now we have to decide the material to be used. For this we selected “Nylon 6” which is light weight and can provide sufficient strength for the gear to work. Moreover it’s a polymer so it won’t be rusting.

21) So from the tensile stress table we find out the ultimate tensile stress from which we will find out the allowable stress that is to be used in lewis equation.

$\Rightarrow$ Ultimate Tensile stress(S_{ut}) of

Nylon 6 = 45-90 MPa = 6.52-13.04 ksi

$\Rightarrow$ Taking avg, S_{ut} = 10.15ksi

[NOTE: As the value of tensile stress varies at different condition so we have taken the average value]

$$\Rightarrow \text{allowable tensile stress}(S_{al}) = S_{ut} / N_{sf} = 10.15 / 1.5 = 6.76 \text{ksi}$$

22) From the Lewis standard table,

$$\Rightarrow \text{Lewis Factor}(Y) = 0.359$$

23) Now putting all the values of variables back in lewis equation and solving it for ‘b’

$$\Rightarrow b = (419.55 * 1.5 * 6.35) / (6.76 * 0.359) = 1.64 \text{in} \approx 40 \text{mm}$$

24) As for checking whether the gear is valid, we can get it if the value of b lies between the below values

$$\Rightarrow \frac{8}{P_d} < b < \frac{12.5}{P_d}$$

$$\Rightarrow 1.25 < b < 1.96 \text{ in}$$

$\Rightarrow$ So as the value $b = 1.64$ lies between the given value therefore gear is valid.

7.1.4 For finding the suitable gear for motor 2:

25) Similarly for finding out the gear for 2nd motor, we will apply the same method. But here the transmitted force will be changed as on the 2nd gear the force applied will only be of 2nd gate.

Again We take a standard gear having No. of teeth = 30 ; Pitch Diameter(D) = 120mm = 4.72in; Module = 4 ; for finding out its face width and verifying that the gear is valid.

26) As we know that the overall force is the transmitted force on the motor so we calculate surface speed and dynamic force acting on the gear

$$\Rightarrow \text{Transmitted force}(F_t) = \text{overall force}(F)$$

$$\Rightarrow F_t = 825 \text{N} = 185.4 \text{lb}$$

$$\Rightarrow \text{Diametral Pitch}(P_d) = \text{No. of teeth} / \text{Pitch diameter}$$

$$\Rightarrow P_d = 30/4.72 = 6.35 \text{ in}$$

$$\Rightarrow \text{Surface Speed } (V_m) = (\pi * D * N) / 12 = (3.14 * 4.72 * 63.69) / 12$$

$$\Rightarrow V_m = 78.7 \text{ ft/min}$$

$$\Rightarrow \text{Dynamic Force } (F_d) = [(600 + V_m) * F_t] / 600 = (678.7 * 185.4) / 600$$

$$\Rightarrow F_d = 209.71 \text{ lb}$$

27) And the relation between F_s and F_d are:

$$\Rightarrow \frac{\text{Allowable force } (F_s)}{\text{Safety Factor } (N_{sf})} \geq \text{Dynamic force } (F_d)$$

28) So the new lewis equation comes out to be

$$\Rightarrow F_d = (S_{al} * b * Y) / (N_{sf} * P_d)$$

29) Therefore the value of face width (b) =

$$\Rightarrow b = \frac{F_d * N_{sf} * P_d}{S_{al} * Y}$$

30) So from the tensile stress table we find out the ultimate tensile stress from which we will find out the allowable stress that is to be used in lewis equation.

$$\Rightarrow \text{Ultimate Tensile stress } (S_{ut}) \text{ of}$$

$$\text{Nylon 6} = 45\text{-}90 \text{ MPa} = 6.52\text{-}13.04 \text{ ksi}$$

$$\Rightarrow \text{Taking min value, } S_{ut} = 6.52 \text{ ksi}$$

[NOTE: As the value of tensile stress varies at different condition so we have taken the average value in previous case but here as the load of only single gate is applying therefore we take the minimum value.]

$$\Rightarrow \text{allowable tensile stress } (S_{al}) = S_{ut} / N_{sf} = 6.52 / 1.5 = 4.34 \text{ ksi}$$

31) From the Lewis standard table,

$$\Rightarrow \text{Lewis Factor } (Y) = 0.359$$

32) Now putting all the values of variables back in lewis equation and solving it for 'b'

$$\Rightarrow b = (209.71 * 1.5 * 6.35) / (4340 * 0.359) = 1.28 \text{ in} \approx 33 \text{ mm}$$

33) As for checking whether the gear is valid, we can get it if the value of b lies between the below values

$$\Rightarrow \frac{8}{P_d} < b < \frac{12.5}{P_d}$$

$$\Rightarrow 1.25 < b < 1.96 \text{ in}$$

$\Rightarrow$ So as the value $b = 1.28$ lies between the given value therefore gear is valid.

7.1.5 For finding the suitable rack:

34) As the material will be the same and the transmitted force will be equal and opposite therefore we have taken the thickness of the rack **40mm** same as of gear.

So the rack would be of module (m) = 4

[Same as gear module for better interaction between them]

Length = 3352.8mm

7.1.6 For finding the Guideway and roller:

- 35) As for finding out the guideway we decide that it should be of stainless steel so that its rust free. Moreover it should be of convex shape so that no debris or stones can be stuck there. As for strength we directly asked the vendor according to its weight carrying capacity. Its overall length is around 22m.
- 36) As for the roller it will also be made of steel and can be asked on the weight carrying capacity from the vendor .moreover its wheel will be in concave shape.

7.1.7 For finding bolts and nuts:

- 37) The bolt and nut that are to be used are of grade 4.8 as tensile strength and yield stress of grade 4.8 bot are:

⇒ Tensile strength=400 MPa

⇒ Yield stress=320 MPa

Which are more than enough to hold the motor on the base and for the support system.